

OODA Loop

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Observe → Orient → Decide → Act, at AI speed

PROFESSIONAL DEFINITION

The OODA Loop is a four-stage decision cycle — Observe, Orient, Decide, Act — that models how an actor adapts to a changing environment by continuously re-cycling through the four stages faster than rivals.

WHO DEVELOPED IT

Developed by United States Air Force colonel and military strategist John Boyd in the 1970s–80s from his study of fighter-pilot decision-making and the dynamics of conflict.

WHY IT WAS DEVELOPED

Boyd sought to explain why outnumbered pilots could prevail: not by superior force, but by completing the decision cycle faster, repeatedly disorienting the opponent.

FIGURE 1 · OODA LOOP

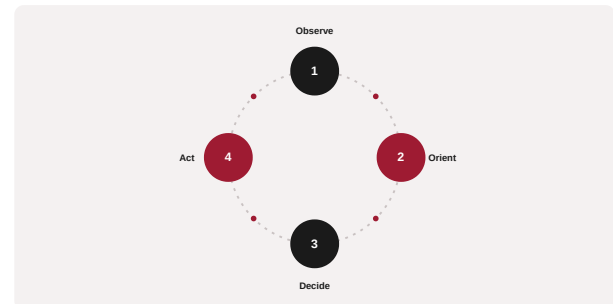


Figure 1. A schematic of the OODA Loop as applied in BARBM312 (illustrative).

HOW IT IS USED

Map the four stages to a real decision process, then look for delays. AI typically compresses Observe and Orient; the discipline is to keep Decide and Act human and fast.

WHEN IT IS USED

In fast-moving competitive or operational contexts — pricing, fraud, content, incident response — where the speed of re-decision matters more than the size of any single decision.

BY WHOM IT IS USED

Operations leaders, strategists, and anyone designing real-time or near-real-time AI decision systems.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Unlike linear plan-then-execute models, OODA is explicitly cyclical and adversarial, treating tempo itself as a source of advantage — a natural fit for AI-accelerated decision loops.

RELATED FRAMEWORKS IN THIS COURSE

DIKW Pyramid · Slibrary 1

Decision-Intelligence Stack · Slibrary 1

Data Value Chain · Slibrary 1

Insight-to-Action Gap · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Boyd, J. (1986) 'Patterns of Conflict' (unpublished briefing).

Osinga, F. (2007) Science, Strategy and War: The Strategic Theory of John Boyd. London: Routledge.